

Newton's Laws of Motion

A person with blonde hair tied back, wearing a grey tank top and sunglasses, is seen from behind while kayaking in a yellow kayak. They are using an orange paddle to propel the kayak forward on a calm, greenish-blue body of water. In the background, there are large, dark limestone karsts rising from the water under a cloudy sky. The scene is peaceful and scenic.

3: Forces Occur in Pairs

NEWTON'S LAWS OF MOTION

3.

"FOR EVERY ACTION,
THERE'S AN EQUAL BUT OPPOSITE REACTION."

What does it mean?

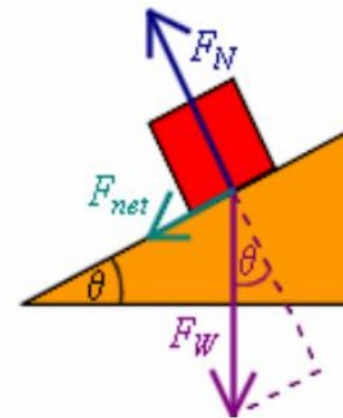
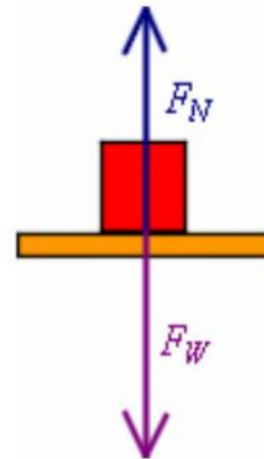
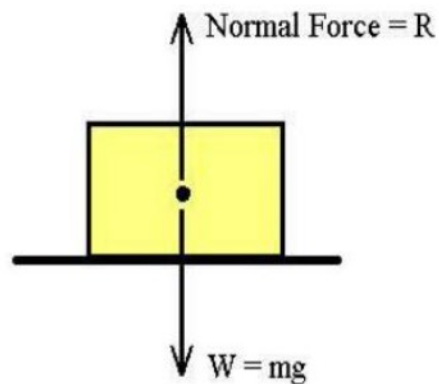
For every force acting on an object, there is an equal force acting in the opposite direction.

Right now, the earth's gravity is pulling you *down* in your seat, but Newton's Third Law says that you are pulling *up* the earth with *equal force*.

Your weight (mass • gravity) is pushing on your chair and the chair is pushing with equal force on you. These forces are balanced.

Normal Force

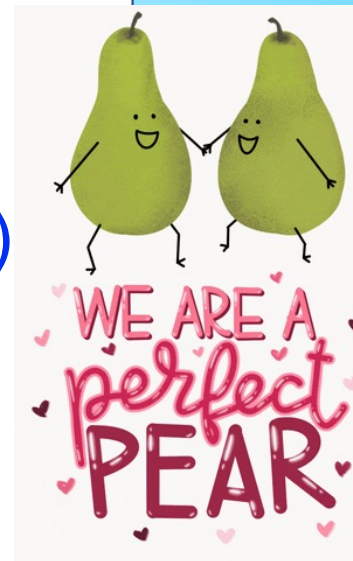
- Normal Force (F_n)- contact force that always acts perpendicular to the surfaces that are touching



Forces always come in pairs

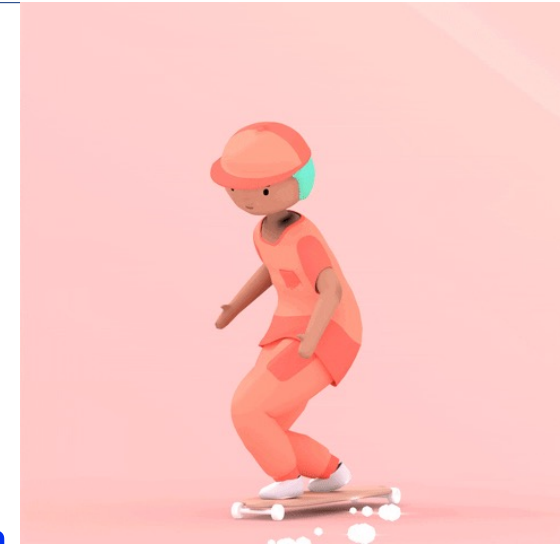
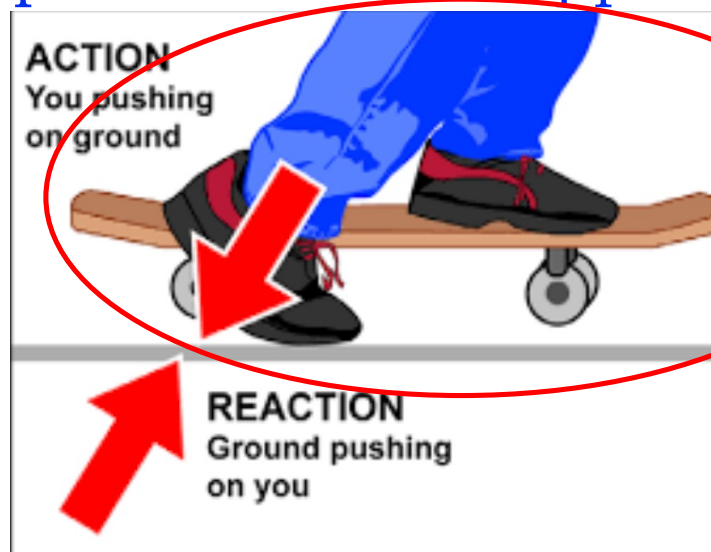
Newton's third law describes the relationship between two forces in an interaction.

- One force is called the **action force**.
- The other force is called the **reaction force**.
- Neither force exists without the other.
- They are equal in strength and opposite in direction.
- They occur at the same time (simultaneously)



Think about it

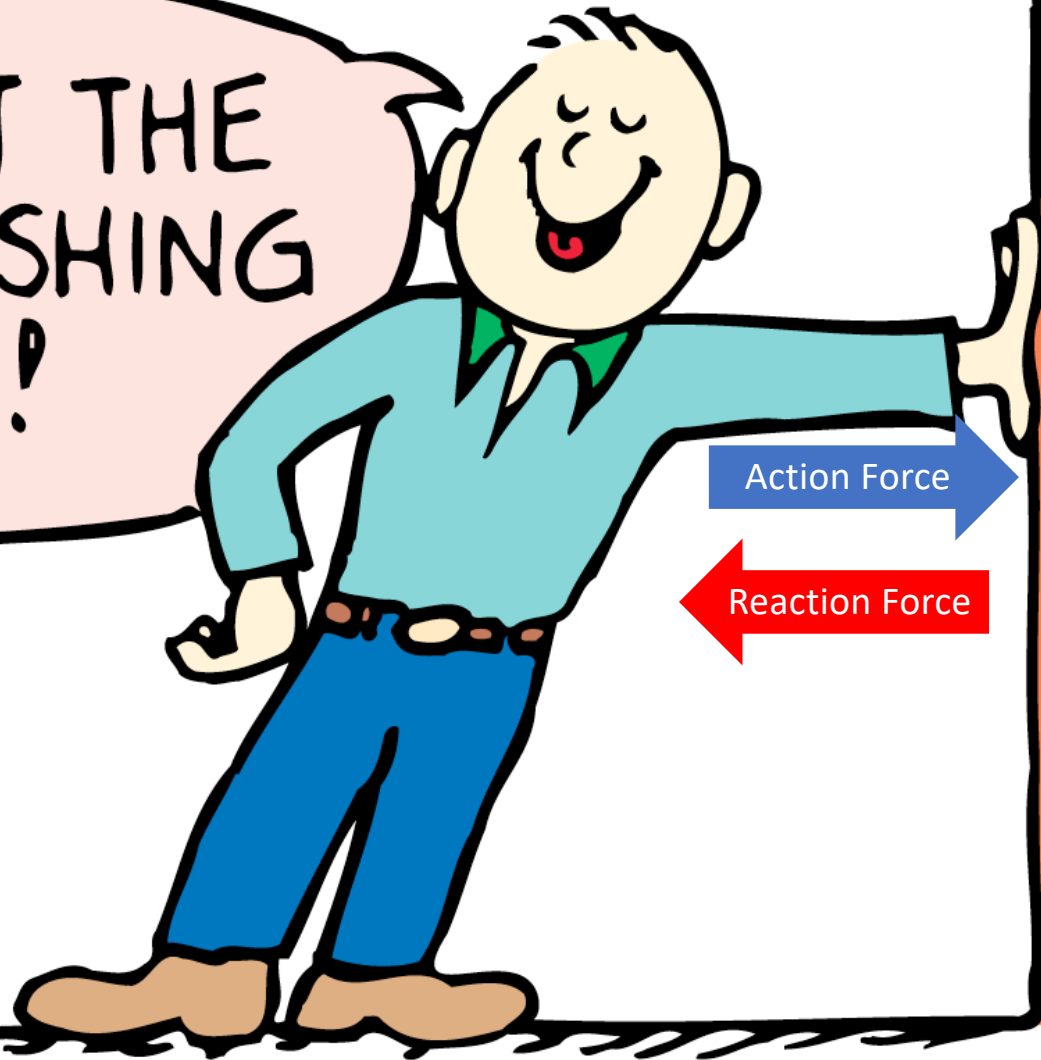
What happens if you are standing on a skateboard and push against the ground? You roll in the opposite direction because you pushed and the ground pushed back on you with equal force in the opposite direction

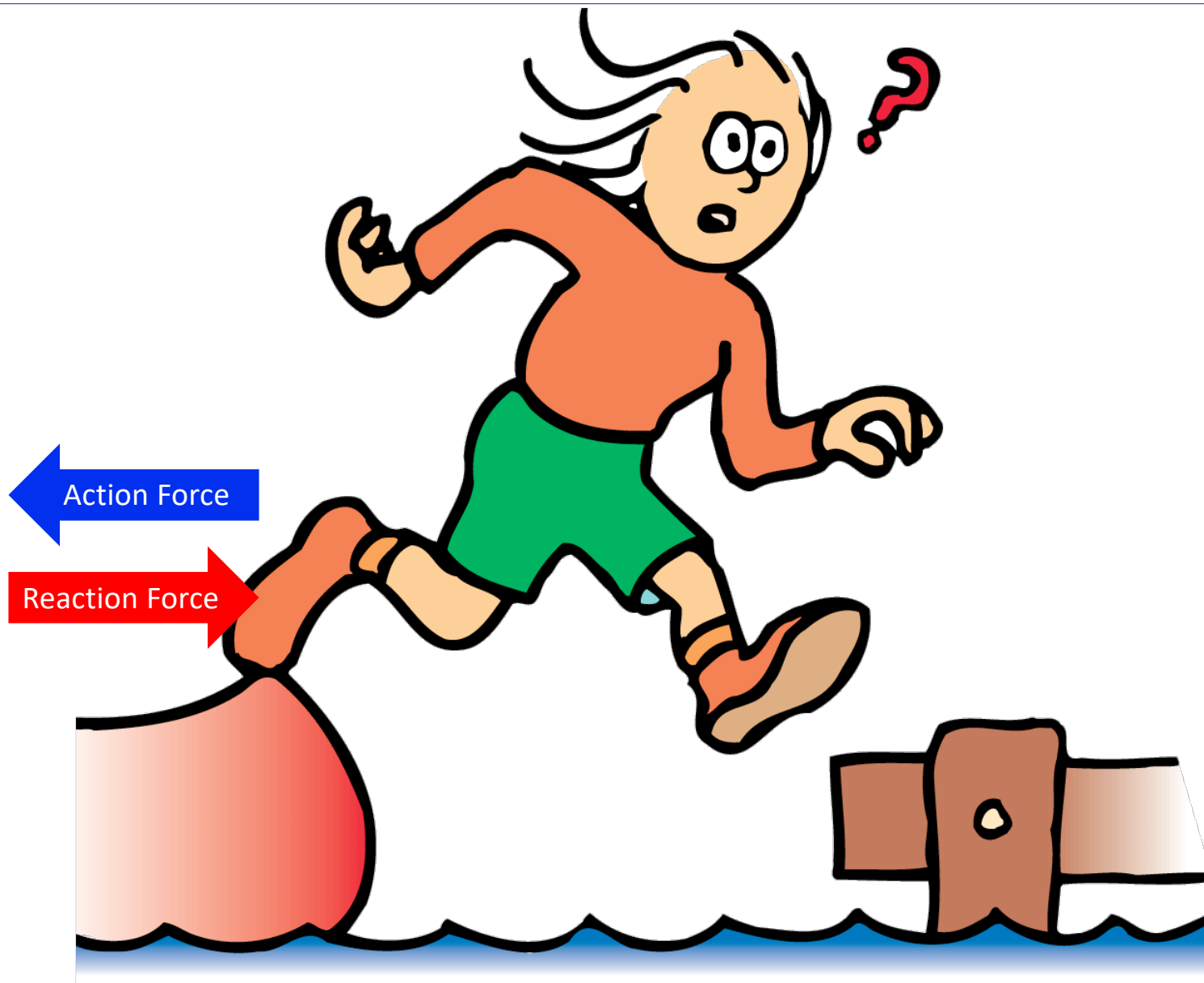


System of Interest

In this example, the person and skateboard are the system we are interested in observing. What other body is interacting with this system?

LOOK AT THE
WALL PUSHING
ON ME!





Think about it

How does Newton's 3rd Law help a fish swim?

How does Newton's 3rd Law help a bird fly?

How does Newton's 3rd Law help a you run?

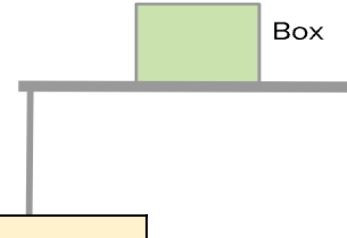
How does Newton's 3rd Law help a rocket launch?

When a rocket launches, which is greater, the action force (the rocket pushing on the Earth) or the reaction force (the Earth pushing on the rocket)?

Which accelerates more, the rocket or the Earth? Why?



Match the action force with its reaction force.



The earth's gravity pulls down on the box

The box's gravity pulls up on the earth

The box's weight pushes down on the table

The normal force from the table pushes up on the box

The table's weight pushes down on the ground

The normal force from the ground pushes up on the table

The earth's gravity pulls down on the table

The table's gravity pulls up on the earth

Will the chair break?

A carpenter makes a chair that can withstand 2,500N of force before it breaks.

A baby elephant with a mass of 200 kg sits on the chair.

The acceleration due to gravity (g) is 10 m/s^2

Let's use what we've learned about Newton's Laws of Motion to find out.

$$1\text{N} = 1\text{Kg} \cdot \frac{m}{s^2}$$

